Smart Detection of Dark Patterns in Web Interfaces Using AI-Based Multi-Agent Systems

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***Abstract*— The dark patterns of interface design, which intentionally exploit cognitive and informational biases and asymmetries, are a new ubiquitous phenomenon in the digital world and increasingly threaten cybersecurity and privacy [1] [2]. This survey uses recent research from 2016-2025 on the definition, taxonomy, and empirical prevalence of dark patterns, with a particular importance on their role in influencing data disclosure, security settings, and account and permission management [3]. Based on interdisciplinary studies in HCI, law, and security, the paper surveys existing taxonomies and ontologies, examines large-scale studies of dark patterns "in the wild," and surveys state-of-the-art detection methods, ranging from rule-based heuristics, machine learning models, to hybrid auditing processes. The survey points out the existing gaps in harmonized taxonomies, representative datasets, and real-time security-aware detection, particularly in mobile, IoT, and platform environments. Based on the survey, the paper argues for a future multi-agent AI system for real-time detection and classification of dark patterns on live webpages, treating interface manipulation as a first-class security issue, alongside traditional technical vulnerabilities [9] [15].**

***Keywords*— Dark patterns, Deceptive design, User interface security, Cybersecurity, Privacy, Human computer interaction, Automated detection, Multi-agent systems.**

I. INTRODUCTION

A. Background: Dark Patterns in Cybersecurity

Over the past decade, online services have increasingly turned to persuasive interface design as a means of encouraging engagement, data collection, and monetization, and in doing so, have increasingly blurred the boundaries between legitimate nudging and outright manipulation [1]. These patterns tend to be subtle when viewed in isolation, such as the pre-selected checkbox, the confusing opt-out link, or the misleading button label, but when viewed in aggregate, can have a profound impact on user behavior, causing it to diverge sharply from informed and privacy-respecting choices[13]. Dark patterns are particularly difficult to identify in the context of security and privacy, as they tend to operate at main points of decision, such as account creation, permission requests, consent to data processing, and security configuration, where users eventually facing complexity and information asymmetry[17].

B. Problem Statement and Motivation

Although it was first considered a matter of ethical design, dark patterns have developed to produce measurable risks to cybersecurity and privacy. [5] [7]. They lead to an excessive amount of data collection, hinder the withdrawal of consent, and preserve insecure defaults. [11]. Regarding technical solutions, there is no standardized framework for automated detection, making it harder to integrate protection into design and audit processes. [17]. This review, therefore, intends to connect theoretical, empirical, and regulatory perspectives to show how dark patterns undermine secure behavior and provide technical guidance for mitigation. [18].

The aim is not only to explain what dark patterns are but to explain how their in-evident secure behavior is performed, why existing protection mechanisms are inadequate, and what technical innovations may help to facilitate more trustworthy digital environments. [5] [7] [17].

C. Research Objectives and Contributions

Dark patterns may vary significantly depending on the context, whether it is web consent banners, mobile permission requests, or larger platform ecosystems [4] [10]. Dark patterns target human users but may also mislead AI agents functioning as human user [9] [16]. The literature review aims to integrate existing taxonomies, approaches to detection, and the role of dark patterns in the context of cybersecurity [8]. It also talks about approaches to create multi-platform detection systems that could safeguard human and AI agents in dynamic digital environments [4] [9].

D. methodological criteria for this study

A systematic literature review was performed to examine the literature on dark patterns and deceptive user interface design. Relevant literature was searched from various academic databases, such as Scopus, IEEE Xplore, ACM Digital Library, arXiv, and MDPI to make sure that the literature is well-covered from a technical, human-computer interaction, legal, and ethical perspective. The keywords used in combination with Boolean operators include "dark patterns," "deceptive user interfaces," "manipulative design," "AI-based detection," and "ethical UI design." A PRISMA-inspired literature screening was performed, and finally, 20 studies were selected for the qualitative analysis.

Inclusion Criteria

1) Peer-reviewed journal articles and full conference papers published recently (2022–2025) and seminal foundational works (2016-2021) that explicitly investigate dark patterns or deceptive design in digital interfaces.

2) Studies that examine dark patterns in contexts relevant to cybersecurity and privacy, such as consent flows, permission dialogs, account management, authentication, or data-sharing interfaces.

3) Empirical, conceptual, or technical works that provide taxonomies, measurements, user studies, legal analyses, or automated detection methods (rule-based, machine-learning, or hybrid) related to dark patterns.

Exclusion Criteria

1) Studies focused solely on marketing effectiveness, engagement optimization, or interface aesthetics without examining security, privacy, or user autonomy implications.

2) Non-peer-reviewed sources such as blogs, opinion pieces, short posters, and workshop summaries without substantive empirical or technical contributions.

3) Work outside the 2016–2025 window, non-English publications, and papers where full text was not accessible or did not provide enough information to extract methods, findings, or design implications.

II. FOUNDATION OF DARK PATTERNS AND CYBERSECURITY

A. Definitions and Taxonomies of Dark Patterns

Dark patterns are designed interface strategies that manipulate or deceive users into actions that are of prime benefit to the service providers, thus separating them from unintentional or poorly designed choices [1]. Dark patterns have been previously classified into families like obstruction, sneaking, and interface interference [3], which were later extended into comprehensive ontologies that associate specific instances of dark patterns (like confirm-shaming and forced continuity) with manipulation goals [2], [14]. These taxonomies allow for systematic comparisons, detection by rule-based systems and AI, and legal interpretations [2], [14]. To better show the association between dark pattern categories and their respective security risks, Table I associates categories of the taxonomy, potential security risks, and requirements for protection.

TABLE I. Mapping of Dark Pattern Taxonomy to Security and Privacy Implications

Dark Pattern Category	**Security Risk**	**Required Protection**
Sneaking	Unauthorized transactions, consent violation	Explicit consent validation
Forced Action	Excessive data collection, profiling	Permission minimization
Obstruction	Prolonged data retention	Regulatory enforcement
Interface Interference	UI-based deception, phishing-like risk	Secure UI guidelines
Social Engineering -based	Behavioral manipulation	User awareness and detection
AI-driven personalizat ion	Profiling, covert manipulation	Explainable AI, audits
Consent fatigue loops	Consent integrity failure	Standardized consent UI

This work bridges dark pattern taxonomies with their security implications, demonstrating that deceptive interface design functions as a socio-technical attack vector affecting privacy, trust, and system security.

B. Cognitive Biases and User Decision-Making

The effectiveness of dark patterns is grounded in well-known cognitive biases and limitations that shape how people make decisions online. Users tend to:

1) Stick with default options, especially when alternatives require extra effort or are hidden in secondary interfaces.

2) Rely on quick heuristics when faced with time pressure, complex terminology, or very long privacy notices.

Designers can exploit these tendencies by making risky or intrusive choices the default, more visually prominent, or less cognitively demanding than safer alternatives. [7, 17].

C. Security and Privacy Implications

Deceptive interfaces redefine security risk by promoting the overcollection of data, the retention of accounts for too long, and the liberal granting of permissions[6] [10]. Even if protective measures are in place, usability-driven deception can compromise their use and efficacy[7]. To accurately assess cybersecurity posture, evaluation must extend to the clarity of protective functionality and design decisions that could deter good practice.

III. TAXONOMY OF DARK PATTERNS RELEVANT TO CYBERSECURITY

A. Classification by Manipulation Goal (data, money, attention)

For cybersecurity oriented analysis, dark patterns can be categorized based on the specific direct security attribute they attack[3]. Data-extraction-oriented dark patterns compromise confidentiality by increasing the volume and sensitivity of user data stored in the system via bundled consents and ambiguous collection requests[6] [13]. Lock-in-oriented dark patterns, which target the enhancement of lock-in and commitment mechanisms, attack integrity by extending data retention and account exposure via hard-to-reach cancellation or revocation of consent paths[7].

B. Classification by Technical Channel (UI, notifications, consent flows)

Dark patterns can be categorized based on the technical channel used for manipulation [4]. At the element and layout-based channel, imbalanced button visibility, confusing labels, or hidden settings decrease the visibility of secure choices, which directly affects the accuracy of user decision-making [8]. At the notification and timing channel, user attention is manipulated by presenting notifications at times of distraction or exhaustion, resulting in rapid and insecure approvals that do not require informed consent [12]. At the flow or journey channel, disjointed multi-screen designs or spread-out privacy switches decrease system integrity by discouraging secure system configurations

C. Contexts: Web, Mobile, Platforms, and Agents

Dark patterns appear in varying forms in different digital contexts, thereby altering the security risk[4] [8]. In web and online commerce, deceptive checkout or consent flows lead to dangerous user practices like storing payment information or sharing data extensively[4]. In mobile and IoT devices, recurring permission requests or limited default configurations lead to exposure of critical hardware and network information, posing a risk to confidentiality and integrity. In a broader platform ecosystem, default-public options and inter-service integrations increase the risk of exposure at a massive scale, thereby introducing systemic risks to privacy and security[8] [9].

IV. DATASETS AND BENCHMARKS USED IN DARK-PATTERN RESEARCH

A. Public Web Datasets and Crawls

A large part of dark pattern research is based on large-scale crawls of commercial websites and services to investigate the appearance of manipulative design patterns “In the wild.” [4]. In these investigations, researchers automatically visit thousands of webpages, usually e-commerce, news, and service sites, and extract HTML, CSS, and screenshots of prominent flows such as sign-up, cookie consent, checkout, and account management interfaces[5]. The crawled pages are then annotated by hand or semi-automatically to indicate the presence of specific categories of dark patterns, creating datasets that enable both prevalence analysis and machine learning model training. [13].

B. Manually Labeled Dark Pattern Corpora

In addition to crawls, some studies have created manually labeled corpora where expert coders or professional annotators have systematically annotated interface elements and flows according to a pre-defined taxonomy[3]. In these corpora, each example, such as a specific dialog, banner, or step in a flow, is labeled with rich annotations that convey both its surface-level appearance (such as pre-ticked checkbox, countdown timer, or misleading hierarchy) and its underlying dark pattern category (such as obstruction, interface interference, or privacy deception)[8]. Some corpora also include contextual metadata such as domain, platform type, and regulatory region, allowing cross-cutting analyses of patterns by industry

C. For Agents and Automated Tools

More recent work has begun to construct benchmarks specifically designed to evaluate automated tools and

intelligent agents navigating dark-pattern-rich environments. [9]. In these settings, agents are tasked with completing realistic goals such as subscribing without unwanted add-ons, configuring privacy-friendly settings, or uninstalling or deleting an account while interacting with interfaces that may or may not contain dark patterns [15]. The benchmarks usually involve instrumented environments that record the agent's behavior, the path of states visited, and the presence or absence of known instances of dark patterns, allowing for the quantitative assessment of detection as well as robustness [9].

V. DETECTION METHODOLOGIES AND TOOLING

A. Rule-Based and Heuristic Approaches

The early studies on dark pattern detection via rule-based/heuristic approaches employed a set of well-defined methodological criteria for the selection of interfaces that included user decision points, consent screens, or transactional flows, but excluded informational screens that were static or irrelevant [1]. Despite these well-defined inclusion and exclusion criteria, such methods have often struggled with generalization over dynamic or localized settings, which has further impeded scalability as dark pattern designs have evolved over time. [20].

B. Machine Learning and NLP/Computer Vision Models

Machine learning-based researches have demonstrated more rigorous methodological filtering in carefully choosing labeled data sets that included proven manipulative and non-manipulative interfaces, often excluding noisy or ambiguous examples to maintain the integrity of the ground truth[8]. These models broadened detection coverage beyond static pages but introduced methodological challenges such as dataset imbalance, limited domain diversity, and unclear exclusion protocols for borderline cases, which reduced reproducibility and transparency. [16].

C. Implications for Defensive Security

Hybrid frameworks combining both rule-based and learning approaches used a mixed methodological pipeline automatic scanning for initial flagging, as well as human validation through consistent inclusion and exclusion checklists [11]. High confidence examples were filtered by automated software, while ambiguous examples were reviewed by human experts, improving system accuracy through feedback [18]. Table II organizes dark pattern detection techniques by methodology, core idea, datasets, and security implications, showing progression from rule-based to fully automated agent approaches.

TABLE II. Dark Pattern Detection Techniques and Security Relevance

Sr no	Methodology	Technique Type	Database	Security Relevance
1	Rule-Based [4], [5], [20]	Uses fixed logic to flag known patterns like pre-ticked boxes, hidden options	Web / Mobile	Detects sneaking and forced actions but is hard to scale and maintain.
2	Machine Learning [8], [12], [16]	Learning deceptive features from labeled data (text or visuals).	Mobile apps / Web ads	Find forced actions and nagging, but needs large, labeled data. interpretability.
3	Computer Vision [11]	Detects visual asymmetries in screenshots and layouts.	UI prototypes / Screenshots	Reduces deceptive defaults but is sensitive to layout changes.
4	Hybrid Models [4], [20]	Combines rules with ML for balanced detection.	Web / Mobile	Detects sneaking and nagging but requires expert tuning.

VI. IDENTIFIED RESEARCH GAPS

A. Limitations of Current Taxonomies and Coverage

However, despite the progress that has been made, the current taxonomies and ontologies of dark patterns are still fragmented and not exhaustive[2].The taxonomies of dark patterns that have been developed by practitioners are more likely to emphasize visual illustrations, whereas the models that have been developed by the fields of law and psychology are more likely to emphasize abstract concepts such as autonomy and coercion[3]. As a result, the current taxonomies tend to be unclear on how to prioritize manipulative patterns based on their potential security effects [3].

B. Gaps in Automated Detection and Real-Time Monitoring

Existing rule-based and machine-learning detectors are useful for static analysis but not working for real-time auditing. Most operate offline on curated datasets, disconnected from live systems or user interactions. This restricts autonomous detection and adaptive defense against evolving manipulative interfaces. Real-time detection faces problems like dynamic content handling, performance optimization, and maintaining user privacy during inspection.

C. *Lack of Labeled Datasets and Cross-Platform Studies*

One of the largest research gaps is in the development of representative datasets that can be used across a variety of platforms, industries, and timescales[8]. This is further exacerbated by the lack of representation of crucial sectors such as healthcare, education, and government services, which hinders the application of these results across the board[16]. Moreover, due to the constantly adaptive nature of modern digital services, static datasets become obsolete almost as soon as they are created[8].

VII. FUTURE DIRECTIONS AND PROPOSED MULTI-AGENT SOLUTION

A. *Requirements for Robust Dark-Pattern Detection Systems*

For a better comprehension of the manipulation process on the new interaction patterns, these systems must be able to process multiple data streams, such as text, image, code markup, and interaction behavior data [9]. By integrating these detection capabilities into design processes, compliance tools, and browser interfaces, we can empower developers and end-users alike to detect and counter manipulative patterns the moment they emerge [9] [15].

B. *Role of Multi-Agent Architectures and Advanced Models*

Employing multi-agent architecture can greatly improve accuracy and efficiency by applying specific roles to a group of cooperative agents [9]. As this architecture is layered, it has the potential to offer a more complex and contextually aware interpretation that is able to distinguish between a dark pattern and a harmless design quirk. This leads to a set of human-readable reports that provide clear and actionable insights for developers, auditors, and regulators alike [9] [15].

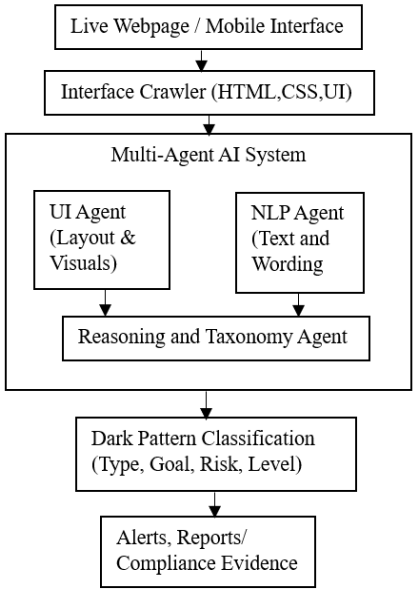

FIGURE1. Multi-Agent Dark Pattern Detection Framework

C. *Alignment with SDGs and Ethical Design Principles*

By incorporating dark pattern detection into ethical frameworks and review cycles, we can start to shift the incentive structure of developers away from exploitative patterns[19]. From a policy perspective, the use of detection methods in digital rights and sustainable development indicates the significance of the methods in developing a fair and transparent digital world [18].

VIII. CONCLUSION

The work reviewed in this paper shows that dark patterns are across web, mobile, IoT, and large platform ecosystems, interfaces are routinely shaped to exploit cognitive biases, defaults, and information asymmetries, nudging users toward excessive data disclosure, weakened security configurations, and long-lived accounts and permissions that expand the overall attack surface[3] [7].

Methodological advances in rule-based detection, machine learning, and hybrid auditing workflows offer promising tools for identifying dark patterns at scale, but most current systems operate offline, are limited to specific contexts, or struggle to deliver explanations that are robust enough for legal and regulatory use[15] [18].

This is exemplified by treating the detection of dark patterns as being as critical to security as other traditional technical controls, and that these should be implemented as part of a defense-in-depth strategy rather than solely viewing the mitigation of dark patterns as an ethical or usability problem. Additionally, the suggested direction of developing a real-time, multi-agent AI system for identifying and classifying dark patterns on live web pages

directly supports this new generation of frameworks by employing multiple agents working together to crawl, observe, classify, and explain dark patterns based on shared reasoning between the various agents regarding structural, linguistic, and user flow attributes.

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